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AI-Powered Automatic Board Cleaner

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ABSTRACT

Maintaining clean whiteboards is crucial for effective communication in educational, corporate, and collaborative settings, yet traditional manual cleaning methods are often labor-intensive, time-consuming, and inconsistent, leaving streaks or residue that compromise visual clarity. Accessibility challenges, such as cleaning boards at varying heights or hard-to-reach locations, further exacerbate the issue, causing physical strain and potential health risks. This project aims to develop an AI-powered automated whiteboard cleaner that employs advanced path optimization algorithms to deliver efficient, consistent, and streak-free cleaning with full surface coverage. Designed for ease of use, the system overcomes accessibility barriers, reduces physical effort, and enhances safety. By integrating cutting-edge technology with practical design, this solution provides a reliable, ergonomic, and time-saving alternative to traditional methods, significantly improving productivity in environments where whiteboard cleaning is essential.

Keywords:- Whiteboard Cleaning, Automation, Arduino, DC Motor Control, Raspberry pi, Open cv, Classroom Technology, Smart Board Cleaning

INTRODUCTION

Whiteboards play an important role in classrooms, corporate offices, and collaborative environments, serving as a primary medium for communication and idea sharing. Despite their usefulness, the process of cleaning them manually is still inefficient and often inconsistent, leading to uneven wiping and visible marker residue. Regular cleaning also becomes physically tiring for users, especially when dealing with large boards or boards placed at heights that require stretching or repetitive movements. In highly active environments, these factors may contribute to reduced hygiene, discomfort, and loss of productivity.

To overcome these limitations, this project presents an AI-based Automatic Board Cleaner, a smart system designed to automate the cleaning process with minimal human intervention. Unlike traditional approaches that rely solely on manual interaction, the proposed design integrates Raspberry Pi, Arduino, computer vision, and a Flutter-based mobile interface to enable intelligent detection of writing, optimized cleaning decisions, and user-friendly controls. The system analyzes board content in real time using

OpenCV and initiates the cleaning mechanism when required, improving energy efficiency and reducing unnecessary motor movement.

The automated wiper mechanism delivers consistent pressure and smooth horizontal wiping, ensuring streak-free cleaning every time. With features such as mobile control, voice commands, and live video monitoring, the system is adaptable to classrooms, offices, seminar halls, and other learning or collaborative spaces. By combining automation, intelligent sensing, and cloud-based communication, this project aims to make board cleaning faster, safer, and more convenient.

LITERATURE SURVEY

In [1] A low-cost automated whiteboard cleaning system was introduced using a brush mounted on a horizontal rail driven by a DC motor. The wiper moves from one end of the board to the other and returns to its initial position after cleaning. The study highlights the reduction in cleaning time and effort in classroom environments. However, the device lacks adaptability for different board sizes and fails to clean selective regions effectively.

In [2] A smart cleaner controlled using **NodeMCU and Wi-Fi connectivity** employed a chain-driven wiper mechanism. The unit was managed through an Android application developed using **MIT App Inventor**, providing directional controls like forward, stop, and backward. The system enables remote interaction, which protects users from exposure to chalk and marker residue. Performance analysis showed considerable improvement in convenience and uniform wiping. Nevertheless, the reliance on Wi-Fi range and instability in chain tensioning affected movement precision, causing motor overload and occasional cleaning gaps on board edges.

In [3] Another study developed a **lead screw-based automated cleaning system** powered by a 12V wiper motor. The wiping pad traveled across a threaded rod, ensuring slow but highly controlled horizontal movement. This approach enhanced durability and consistency, making it suitable for schools requiring frequent board usage. Quantitative testing revealed superior erasing speed compared to hand wiping. However, long-term evaluations indicated that thread wear reduces motion smoothness, and lack of selective region cleaning remains a drawback, making the system less flexible for quick mid-lecture corrections.

In [4] A semi-automated smart cleaner employed a **Double Pole Double Throw**

(DPDT) switch and linear rail movement to reduce manual fatigue. The cleaning action could be activated simply by pressing a switch, eliminating the need to rub the board manually. Its simplicity makes it feasible in low-budget institutions. Laboratory tests confirmed stable wiping performance and reduced physical strain for teachers. Nonetheless, the system depends entirely on user interaction and lacks automation intelligence, remote control, or sensing capabilities, restricting its evolution into smart classroom ecosystems.

In [5] The paper introduces a microcontroller-based cleaning solution used **Arduino UNO and L293D driver** to drive multiple DC motors for quick cleaning. The system reduces chalk dust exposure and enhances time efficiency. Nevertheless, it offers limited movement precision due to unidirectional wiping and depends heavily on external power supply, making it less portable.

In [6] Another approach integrated an **automated vacuum-based dust absorption system** with a motorized board cleaner. Its key advantage lies in removing chalk dust, which is hazardous in classrooms. Laboratory results demonstrated improved hygiene conditions and reduced lung irritation for teachers exposed to chalk. However, the device required regular dust bag maintenance and frequent filter replacement. The vacuum motor added noise and cost to the system, making it less suitable for marker-based whiteboards where dust collection is not essential.

In [7] The paper proposes A **rack-and-pinion smart cleaner** was introduced to achieve uniform wiping pressure across the board. The motor transferred torque through gears, ensuring steady mechanical motion. Tests indicated superior cleaning quality due to even pressure. While this improved wiping precision, maintenance challenges increased, as frequent lubrication, gear wear, and higher power

consumption were reported. The system also required professional installation, limiting its viability in low-resource classrooms.

In [8] A **belt-driven mechanism** was proposed to reduce mechanical complexity and cost. The lightweight build reduced load on the DC motor, improving motion efficiency. Initial evaluations highlighted a reduction in manufacturing cost and smooth cleaning movement. However, long-term testing indicated belt slackness, requiring periodic replacement or tightening. This caused uneven wiping over time, especially in high-humidity environments where belt elasticity changed.

In [9] The paper Some designs incorporated Bluetooth-controlled remote wiping, offering teachers distance-based operation to avoid interrupting lectures. Wireless activation increased classroom comfort and reduced cleaning time. However, Bluetooth interference in crowded environments and limited operating range (typically 8–12 meters) affected reliability. Moreover, the system required pairing each time the board was to be cleaned, reducing operational convenience during rapid lecture transitions.

In [10] The paper evaluates A design leveraging servo motors for precision wiping enabled controlled motion in both horizontal and partial cleaning operations. Servo control improved adaptability for selective wiping, ideal for partial erasing. Results showed higher precision, especially for fixed marker stains. Nevertheless, servo motors required more complex calibration and added cost. Frequent torque fluctuations caused overheating, complicating implementation for large boards and restricting cost-effective scalability.

In [11] The paper presents a dual-wiper mechanism using spring-loaded pads improved friction control for removing dried markers. The spring action ensured

consistent contact pressure across the whiteboard surface, achieving better wiping performance. Experimental studies revealed increased efficiency in cleaning stubborn residue. However, the

mechanism required periodic spring replacement, and tension imbalance resulted in uneven cleaning after prolonged usage, limiting long-term reliability

Authors	Title	Research Focus	Remarks
Romeo S. Fono-Tamo et al., 2021	Performance Evaluation of an Automated Whiteboard Cleaner	Full mechanical evaluation of board cleaner using DC motor & sliding rail.	Lacks intelligent control, precision depends on component quality
Dr. N. Gayathri et al., 2023	Design and Fabrication of Automated Board Cleaner	Lead screw-based design for faster wiping	No selective cleaning & pad pressure inconsistency
L. Akpeti, 2023	Upgraded Automatic Whiteboard Cleaning System	Introduces partial cleaning & residue reduction	Highly complex setup needs calibration

Shazleen Azwa K. & S. Zaharah, 2024	Easy-Erase Whiteboard Prototype	Arduino + cleaning fluid system	Maintenance high due to fluid dependency
IJIRT, 2024	Embedded Based Automatic Board Clean Robot	Arduino + servo + vacuum - assisted cleaning	Noise, power usage & vacuum maintenance issues
Srija Reddy Thummala pally, 2022	Automatic Whiteboard Cleaner Using Arduino	Uses rack-and-pinion motor mechanism	Slider wear over time reduces efficiency.
IJCRT, 2024	Automatic Blackboard Writing and Cleaning System	Dual mode: writing + cleaning using stepper & servo	Complex configuration increases cost & calibration

Arxiv.org Study, 2023	Implementation of Arduino in Science : Bibliometric Analysis	Trends & justification for Arduino in automation	No specific cleaning design, but supports tech selection	IRJET, 2023	Motorized Smart Wiper System for Teaching Boards	DC motor-based automatic wiping with timing control	Limited adaptability for board size variation .
Lokesh Kumar et al., 2021	Smart Classroom Using Arduino and IoT	IoT devices for classroom automation	Doesn't integrate cleaning, but justifies smart board systems				
IJEAT, 2022	Smart Dust-Free Blackboard System	Chalk dust removal using suction motors & filters	High maintenance & cost due to filter replacements.				

IMPLEMENTATION

From the Figure 1, This image illustrates the architecture of a smart board barrier system that integrates user interaction, cloud-based data management, and hardware automation. At the front end, users interact with the system through a mobile or web application, which serves as the interface for sending commands or receiving updates. These interactions are routed through Firebase, a cloud database that acts as the central communication hub. Firebase stores user requests and system status, enabling real-time data exchange between the interface and the hardware. On the hardware side, the

system includes a microcontroller (such as a Raspberry Pi), an Arduino board, a camera module, and a motor. The microcontroller connects to Firebase to retrieve commands and passes them to the Arduino, which controls the motor to physically open or close the barrier. The camera module may be used for visual verification or monitoring, sending feedback back to Firebase or the user interface. Overall, the architecture demonstrates a seamless loop of interaction where user inputs trigger automated physical responses, supported by cloud-based synchronization and embedded hardware control.

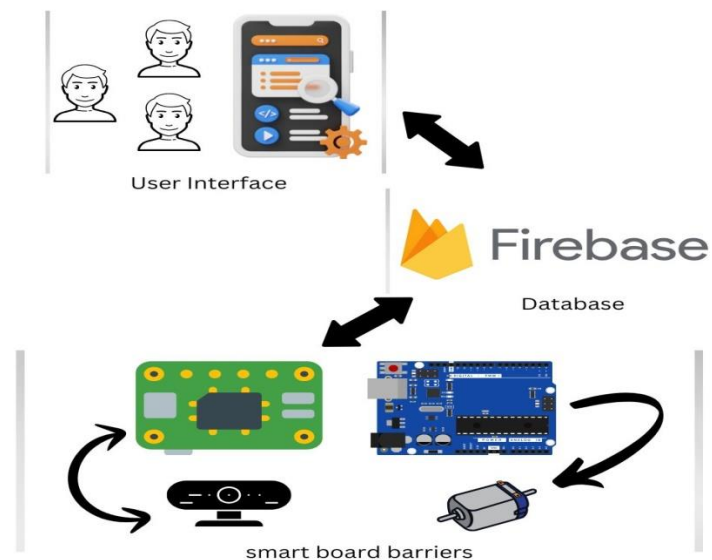


Fig.1:-System design Flowchart

System Description: Modes, Hardware and Data Flow: The proposed AI-Powered Automatic Board Cleaning System is designed to automatically clean a whiteboard using a motorized mechanism controlled by Arduino, and cleaning decisions are taken based on board fill analysis using Raspberry Pi and OpenCV. The system supports three modes—Automatic mode, Manual mode and Voice-controlled mode—and users can operate it using a Flutter mobile application. The system also provides real-time live video streaming of the board through a camera connected to the Raspberry Pi.

Board Fill Analysis and Pre-Processing: To detect whether the board requires cleaning, the Raspberry Pi continuously captures video frames through a webcam and performs image processing using OpenCV.

Frame Capture: Video frames are captured from the camera using `cv2.VideoCapture(0)`.

Grayscale Conversion: The captured frames are converted into grayscale using `cv2.cvtColor()`, allowing easier extraction of writing content.

Binary Thresholding: The grayscale frame is transformed into a binary image (black & white text separation), helping to measure written content.

Fill Percentage Calculation: The white pixel count is obtained using `cv2.countNonZero()`, which calculates the percentage of writing on the board. If the calculated percentage exceeds a threshold (example: 60%), a cleaning request is sent to Firebase.

Firestore Data Flow for Mode Control: The Flutter application allows users to select cleaning modes. Based on user input or automatic triggers, values are updated in the Firestore database.

- **Automatic Mode:** Raspberry Pi updates board fill percentage and triggers cleaning automatically in Firestore.
- **Manual Mode:** User directly presses the start/stop button in the app, and Firestore pushes commands to Arduino.
- **Voice Mode:** Voice commands are converted to text and written to Firestore for execution.

Firestore ensures real-time communication between the mobile App, Raspberry Pi, and Arduino.

Motor Control and Cleaning Mechanism Using Arduino: Arduino controls the servo or DC motor responsible for moving the cleaning wiper horizontally across the board.

- A function `_toggleCleaning()` manages the cleaning sequence by reading Firebase values like `boardStatus/status` and `manual_cleaning`.
 - When the value is **1**, the motor starts wiping.
 - When the value becomes **0**, the motor automatically stops.
- This ensures that the motor operates safely and avoids over-usage during idle periods.

System Training or Configuration: The system does not require machine learning training. Instead, fixed thresholds like board fill percentage, motor speed, and voice command mapping are stored in Firebase. These configurations help the system make accurate and real-time cleaning decisions.

1. Cleaning Execution: When a cleaning request is triggered via app, voice, or automatic detection, Arduino activates the motor to start wiping. The wiper moves horizontally across the board, and its status is updated in Firebase. Once cleaning is complete, the system resets the values to avoid repeated actions.

2. Extracting Filled Board Frames: The Raspberry Pi highlights frames of the board when it detects a large amount of writing. These frames are processed in binary and original formats and streamed to the user. This helps verify whether cleaning was required.

3. Image Streaming Enhancement: For better visibility during monitoring, image enhancement techniques like brightness correction, sharpening and contrast adjustment are applied. These improvements make writing on the board clearer. Enhanced visuals are shown in the app and live stream.

4. Notification & Status System:

After cleaning, the system sends real-time updates such as cleaning completion and board status through Firebase. These notifications are displayed in the mobile app. Future upgrades can include WhatsApp, SMS, or audio alerts.

RESULT AND ANALYSIS

The prototype successfully cleans a standard classroom whiteboard in **6–10 seconds**, compared to manual cleaning which may take **20–30 seconds**. The wiping is uniform without leaving streaks. The device is cost-effective, portable, safe, and easy to operate. It allows teachers to focus more on teaching rather than board cleaning.

CONCLUSION

In this project, a Smart Board Cleaner is developed to automate the cleaning of traditional classroom blackboards in schools, colleges, and training institutes. The system seamlessly integrates a Raspberry Pi, webcam, Arduino, and continuous-rotation motors to eliminate manual cleaning effort and significantly reduce turnaround time between classes. Using real-time image processing, the webcam monitors the chalk fill level on the board, and the Raspberry Pi intelligently triggers the motorized wiping mechanism when cleaning is required. A dedicated Flutter mobile application allows users to control the system in both manual and automatic modes, while Firebase enables real-time video streaming, remote monitoring, and secure data logging. By minimizing chalk dust exposure and reducing manual intervention, the Smart Board Cleaner enhances classroom hygiene, improves operational efficiency, and supports a healthier, more technology-driven learning environment.

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